(2) $a^2 - b^2 \longrightarrow n$ (odd)

$$n = 2k+1 = \underbrace{k}_{b} + \underbrace{(k+1)}_{a}$$

$$\begin{aligned} a^2 - b^2 &= (a+b)(a-b) \\ &= ((k+1)+k)((k+1)-k) \\ &= (2k+1)(\cancel{k}-\cancel{k}+1) \\ &= (2k+1)\cdot 1 = 2k+1 = n \text{ (odd)} \end{aligned}$$

OR

$$\begin{aligned} a^2 - b^2 &= (k+1)^2 - k^2 \\ &= (k^2+2k+1) - k^2 \\ &= \cancel{k^2} - \cancel{k^2} + 2k+1 \\ &= 2k+1 = n \text{ (odd)} \end{aligned}$$

(3) $\underset{(r)}{n} + \underset{(irr)}{k} \rightarrow \underset{(irr)}{m}$

We assume, $\underset{(r)}{n} + \underset{(irr)}{k} \rightarrow \underset{(r)}{m} \Rightarrow \underset{\cancel{(irr)}\ (r)}{k} \rightarrow \underset{(r)}{m} - \underset{(r)}{n}$

$m = \frac{a}{b \neq 0}$, $n = \frac{c}{d \neq 0}$, $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd} = \frac{u}{v}$

$(m-n)+n \rightarrow m \rightarrow$ rational

(4)

1) proof $\sqrt{2}$ is irrational

$\frac{a}{b \neq 0}$ (irreducible)

$\sqrt{2}$ is irrational

$\left(\frac{a}{b}\right)^2 = \left(\sqrt{2}\right)^2$

$\frac{\text{even}}{\text{even}}$ (reducible)

$\frac{a^2}{b^2} = 2$

$2b^2 = a^2$ (a is even)

$2b^2 = (2k)^2 \Rightarrow 2b^2 = 4k^2 \Rightarrow b^2 = 2k^2 \Rightarrow b$ (even)

2) disprove irr * irr → irr

$\sqrt{2} * \sqrt{2} = \underline{2}$ (rational)

⑤ $x + y \geq 2 \longrightarrow x \geq 1$ or $y \geq 1$

$\neg(x \geq 1$ or $y \geq 1) \longrightarrow \neg(x + y \geq 2)$

$x < 1$ and $y < 1 \longrightarrow \underline{x + y < 2}$

⑥ a) if $\underline{n^3+5}$ (odd) $\longrightarrow \underline{n}$ (even)

by contraposition,

$\underline{n}$ (odd) $\longrightarrow \underline{n^3+5}$ (even)

$2k+1 \longrightarrow (2k+1)^3+5$

$(2k+1)(2k+1)(2k+1)+5$

$(8k^3+12k^2+6k+1)+5$

$\underline{2(4k^3+6k^2+3k+3)}$

$2r$ (even)

b)

by contradiction,

$\neg(n^3+5 \rightarrow n)$

$\neg(\neg(n^3+5) \vee n)$

$\underset{\text{(odd)}}{\underline{n^3+5}} \wedge \underset{\text{(odd)}}{\neg n}$

$2k+1$

$(2k+1)^3+5$

$2(\underbrace{4k^3+6k^2+3k+3}_{r})$

$2r$ (even) $\neq$ odd

⑦ $\forall x \, \exists M, \quad \underline{S(B,x)} \longleftrightarrow \underline{\neg S(x,x)}$

barber shaves a man iff man doesn't shave himself

$\underline{S(B,B)} \longleftrightarrow \underline{\neg S(B,B)}$ $\Rightarrow$

barber shaves himself iff doesn't shave himself

P	$\longleftrightarrow \neg P$	
T	F	F
F	T	F